

Overfitting

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can compare training and validation evidence for model capacity.

I can explain overfitting without assuming every gap proves it.

I can use a declared selection rule instead of perfect training fit.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

An overfit model captures training-specific noise, producing a large gap between training and unseen performance.



LEARNING TARGET A has training 98%, validation 58%; B has training 86%, validation 81%.
Compute gaps and select a candidate with a qualification.

Builder name and experiment date:

Readiness for the connected explanation

Can a zero training error guarantee good validation performance?

This week's end goal

By the end, I can explain overfitting using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use overfitting to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A more flexible model can represent complicated relationships and accidental details. Capacity can help when there is enough relevant evidence, but perfect training performance is not the goal. Compare training and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines overfitting and model capacity.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

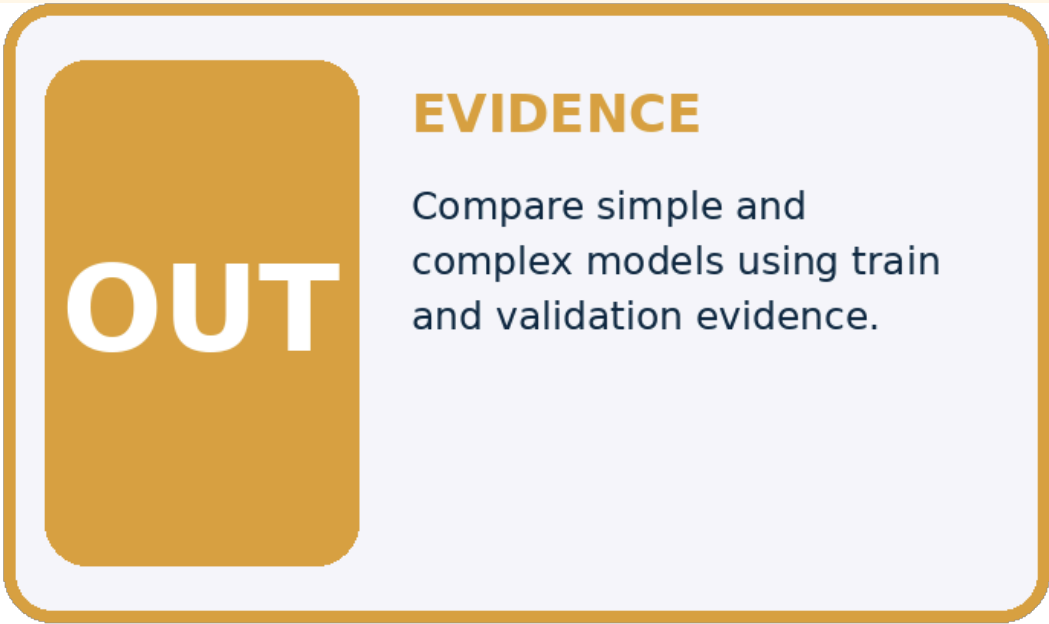
Compare simple and complex models using train and validation evidence.



ML

MODEL

An overfit model captures training-specific noise, producing a large gap between training and unseen performance.



OUT

EVIDENCE

Compare simple and complex models using train and validation evidence.

QUESTION Compare simple and complex models using train and validation evidence.

MODEL IDEA An overfit model captures training-specific noise, producing a large gap between training and unseen performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Simple tree: train 82%, validation 78%; gap 4 points. Deep tree: train 100%, validation 65%; gap 35 points.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

On the same regression split, a constant has training MAE 5 and validation 6, a line has training 2 and validation 3, and a highly flexible curve has training 0 and validation 8. Under smallest validation MAE, choose the line. The curve has the lowest training loss but the worst validation result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Large's validation accuracy rises to 82% after a development revision.

Prediction: Its gap is 18 percentage points; it now beats Small on this validation set.

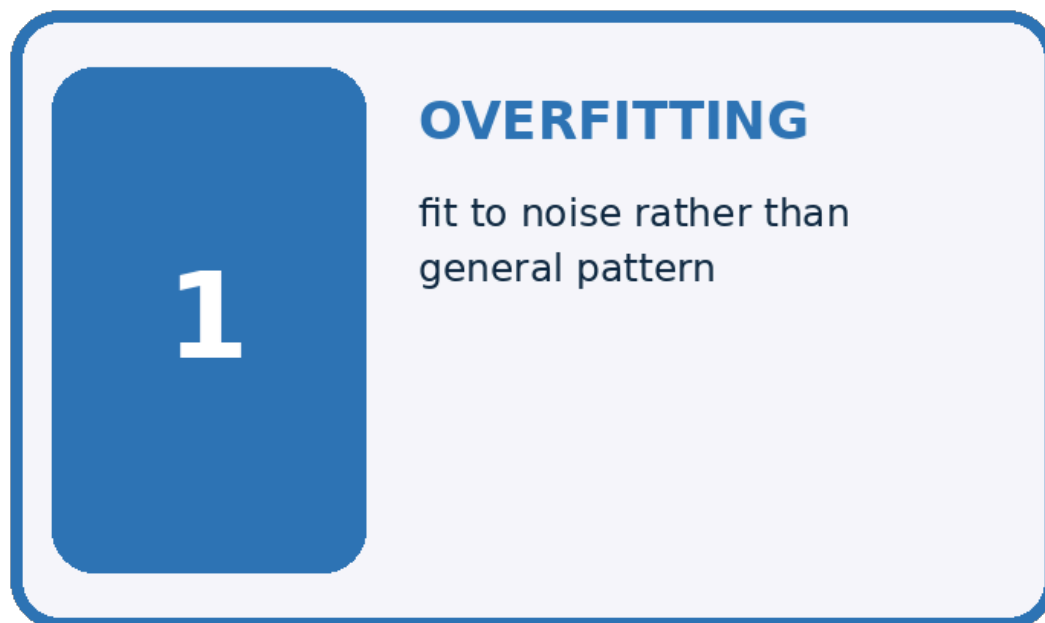
Reason: Validation performance improved, but the training-validation gap is not a complete release rule.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

COMPLEXITY

model capacity to form
detailed boundaries

3

GENERALIZATION GAP

train score minus unseen
score

4

REGULARIZATION

constraint discouraging
excessive complexity

Term · Precise meaning · My ML example

Term: overfitting

Precise meaning: fit to noise rather than general pattern

My ML example: _____

Term: complexity

Precise meaning: model capacity to form detailed boundaries

My ML example: _____

Term: generalization gap

Precise meaning: train score minus unseen score

My ML example: _____

Term: regularization

Precise meaning: constraint discouraging excessive complexity

My ML example: _____

MISCONCEPTION TO REPAIR The model with the highest training accuracy is the best model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Capacity

Meaning: Flexibility of a model to represent patterns

Word: Overfitting

Meaning: Learning details that fail to generalize well

Word: Underfitting

Meaning: Failing to capture useful structure

Word: Generalization

Meaning: Performance on relevant unseen examples

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 27 • WORKED EVIDENCE

- 1 Simple tree: train 82%, validation 78%; gap 4 points.
- 2 Deep tree: train 100%, validation 65%; gap 35 points.
- 3 Deep tree's perfect training result is weaker unseen evidence.
- 4 Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

STEP 1 Simple tree: train 82%, validation 78%; gap 4 points.

STEP 2 Deep tree: train 100%, validation 65%; gap 35 points.

STEP 3 Deep tree's perfect training result is weaker unseen evidence.

STEP 4 Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Overfitting and model capacity at a glance

Part · Evidence or action · Interpretation

Part: Constant

Evidence or action: Training 5

Interpretation: Validation 6

Part: Line

Evidence or action: Training 2

Interpretation: Validation 3

Part: Flexible curve

Evidence or action: Training 0

Interpretation: Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Overfitting

See overfitting as a gap requiring investigation, not a label based only on model size.

The facts we will use

Toy model Small: training 85%, validation 80%. Large: training 100%, validation 60%. Same split and scoring rule.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Training and validation accuracy

percent



Bar length shows the supplied value; the marked baseline is zero.

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to 82% after a development revision.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute gaps

What happens and why: Small gap=5 percentage points; Large gap=40 percentage points.

Step: 2. Select development candidate

What happens and why: Small has the higher validation accuracy despite its lower training fit.

Step: 3. Investigate large model

What happens and why: Check capacity, training choices and data mismatch without tuning on final cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Check capacity, training choices and data mismatch without tuning on final cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A gap can also reflect data shift or sampling noise. Inspect evidence before attributing every gap to capacity.

Read the labelled picture: Does 100% training accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Large's validation accuracy rises to 82% after a development revision. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model with the highest training accuracy is the best model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE The model with the highest training accuracy is the best model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid evidence under the declared rule, keep that evidence visible. Model capacity is one part of an experiment, alongside features, sample size, label quality and the split.

What extra evidence would you inspect before concluding a model memorized noise?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on training and compare on validation. Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Practise with less help: Interpret a gap

Use workbook W2 for the connected example “Overfitting and model capacity”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does 100% training accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Complexity Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.

1. Write the hypothesis and expected evidence before running.
2. Complexity Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: overfitting, complexity, generalization gap, regularization.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The model with the highest training accuracy is the best model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Check Plant Classifier train-validation gaps and limit complexity before final test release.

THIS WEEK'S CHECKPOINT Check Plant Classifier train-validation gaps and limit complexity before final test release.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Compare simple and complex models using train and validation evidence.

REQUIRED PROOF Compare simple and complex models using train and validation evidence.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Learning Curves.

13. Study a complete case

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A scores 100% train, 60% validation; B scores 85%, 80% on the same sets.

The question

Compute gaps and select by validation accuracy.

Worked explanation

A gap 40 percentage points; B gap 5. B has better shared validation accuracy. A suggests overfitting, but investigate distribution differences too.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare training and shared validation

Evidence or stage · Value or result

Evidence or stage: A training

Value or result: 100

Evidence or stage: A validation

Value or result: 60

Evidence or stage: B training

Value or result: 85

Evidence or stage: B validation

Value or result: 80

A scores 100% train, 60% validation; B scores 85%, 80% on the same sets.

Compute gaps and select by validation accuracy.

A gap 40 percentage points; B gap 5. B has better shared validation accuracy. A suggests overfitting, but investigate distribution differences too.

14. Try a new case independently

Independent case: Changed independent case

Model A training 96%, validation 62%; model B training 88%, validation 79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Model A has training accuracy 99% and validation accuracy 70%; B has 88% and 84%. Compute each gap and discuss the evidence.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Check Plant Classifier train-validation gaps and limit complexity before final test release.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

A has training 98%, validation 58%; B has training 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: See overfitting as a gap requiring investigation, not a label based only on model size.

By the end, I can · How I will show it

By the end, I can: Compare training and validation evidence for model capacity

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain overfitting without assuming every gap proves it

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Use a declared selection rule instead of perfect training fit

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Compare training and validation evidence for model capacity

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See overfitting as a gap requiring investigation, not a label based only on model size.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check capacity, training choices and data mismatch without tuning on final cases.

1. Select development candidate
2. Compute gaps
3. Investigate large model

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Investigate large model Check capacity, training choices and data mismatch without tuning on final cases. Read the picture in order. Check capacity, training choices and data mismatch without tuning on final cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Large's validation accuracy rises to 82% after a development revision. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Its gap is 18 percentage points; it now beats Small on this validation set.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set. Validation performance improved, but the training-validation gap is not a complete release rule.

Repair a missing building block

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you

changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.